Security and safety practices

Keep policy text contained and the system safe.

For: Everyone

One chapter of the handbook. The full handbook, with every chapter, is handbook.pdf in the same folder.

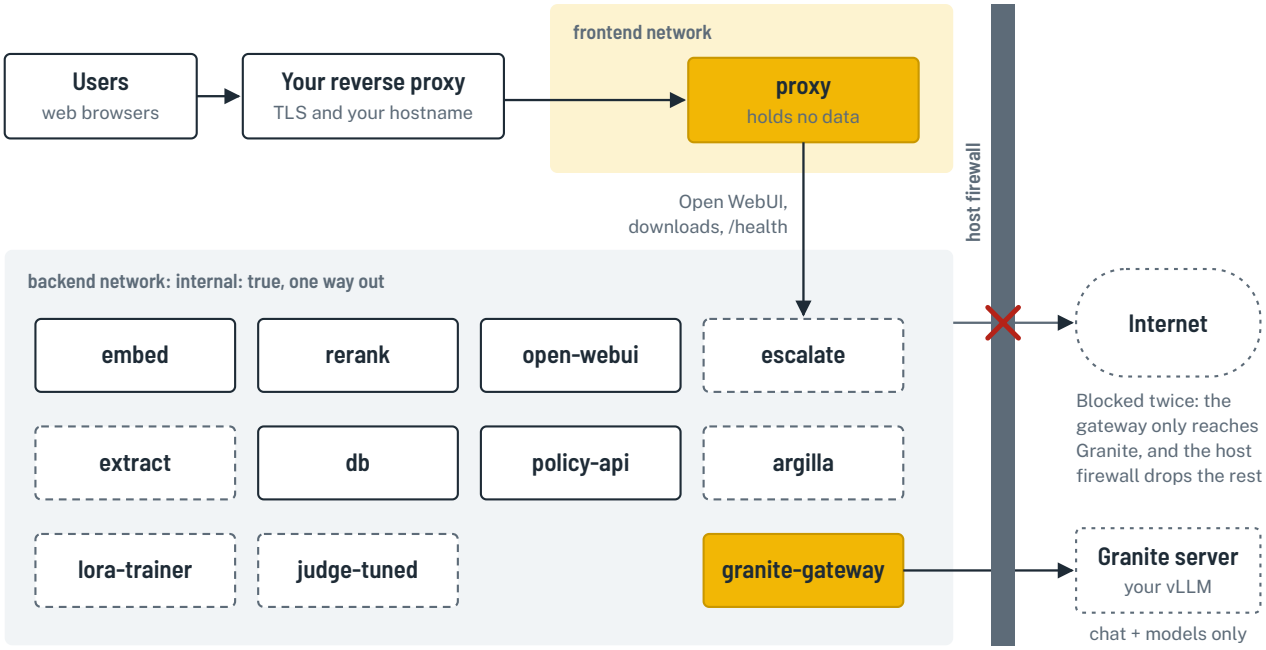
Two things must hold: **policy text never leaves the contained environment**, and **no one acts on an automated decision without a person checking it**. This page explains how the system enforces both, and what you still have to do.

Threats and controls

Risk	Control in this repo	Your job
Policy text leaves the environment	Every service holding text is on the `internal: true` network; the only way out is `granite-gateway`, which passes just chat and model-list calls to your own Granite server; `HF_HUB_OFFLINE`, telemetry off	Run `check_containment.sh`; block egress at the host firewall too, and allow `llm-egress` to reach only the Granite address
Unvetted or tampered model weights	Weights mounted read-only; SHA-256 manifest	Record hashes at download; check before each start; approved-model list
Someone calls the API directly	Bearer key on every route; proxy exposes only downloads and `/health`	Keep `.env` at `chmod 600`; rotate keys
A malicious web page uses a browser to reach the API (DNS rebinding)	`TrustedHostMiddleware` refuses unknown `Host` headers	Keep `ALLOWED_HOSTS` short
A download link is shared or guessed	HMAC-signed links that expire after 72 h; run IDs are random UUIDs	Shorten `LINK_TTL_HOURS` if needed
Instructions hidden inside a policy steer the model ("ignore the above...")	Policy text is fenced as data in every prompt, and fence markers inside it are replaced so it can't break out; the model returns only a fixed JSON schema for each verdict; a "yes" whose quoted evidence isn't really in both texts is downgraded to Maybe for a person to check; the chat model is told never to follow text in documents	Investigate verdicts whose reasons look odd
Policy text becomes a live Excel formula	Cells that would be formulas are forced to text	Nothing
Document or model text adds a link or image to the chat answer (an image URL could carry data out)	Markdown characters in every document and model value are escaped; only the download link is live	Keep Open WebUI on the internal network; block egress so a stray image can't load

Risk	Control in this repo	Your job
A small upload expands into a huge one (zip bomb, giant PDF) or a crafted line slows parsing	Uploads capped at `MAX_UPLOAD_MB`; .docx refused if it unpacks past 50 MB; PDFs capped at 1,000 pages; text capped at 5 M characters; each upload is parsed in its own process (1 GB memory cap, killed after 60 s, at most 2 at once); heading regex can't backtrack	Nothing
The model is wrong	Evidence quotes and reasons on every row; "Maybe" for uncertainty; full audit trail	Keep a person approving every change (below)
Lost history	Every model decision stored in `judgements` with model name and prompt hash	Back up the database (operations.md)

Network containment



Two barriers keep policy text in: a network with one narrow way out, and your host firewall.

- `backend` is a Docker network with `internal: true`, so its containers have no gateway out.
- The backend has exactly **one** way out: `granite-gateway`. It also joins `llm-egress`, a normal Docker network, and forwards to `GRANITE_URL` only. It passes `/v1/chat/completions` and `/v1/models` and nothing else, and checks the server's TLS certificate (`GRANITE_CA_PATH` for a private CA). No other container joins `llm-egress`.
- The Granite API key is held only by the gateway. It adds the key itself and drops any key the caller sent, so Open WebUI and policy-api never see it.
- **Firewall rule to add:** on the host, allow traffic from the `llm-egress` network to the one address and port of your Granite server, and drop everything else from it. Without this rule, the gateway is still limited by its config, but the host would not enforce it.
- Only `proxy` joins `frontend`. nginx serves Open WebUI and only `/exports/<run-id>.xlsx` and `/health` from policy-api. All other API routes return 404 from outside.

- **Don't rely on Docker settings alone.** Also block outbound traffic from the host at your firewall (for example, rules in the `DOCKER-USER` chain on Linux), and keep the server on a network segment without internet access.

Proving it

```
scripts/check_containment.sh
```

The script first checks that `granite-gateway` is the only container on `llm-egress`. Then, for each running service, the script first connects to an internal service. That's the positive control: it proves the probe itself works. Then it tries to reach the internet, and that must fail. The script also fails if it checked nothing. Run it after setup, after every upgrade, and after any network change. Keep the output with your change records.

During development, a network setting meant to block outbound traffic ("no outbound NAT") was ignored by Docker Desktop. The script caught it. That's why containment relies on `internal: true` networks, not driver options.

Secrets

- Every secret lives in `.env` (git-ignored). Generate each with `openssl rand -hex 32`. policy-api refuses to start if its keys are shorter than 32 characters.
- `chmod 600 .env`. Only administrators can read the server.
- **Rotate** by changing the value and restarting the affected services:
- `POLICY_API_KEY`: restart `policy-api` and update the tool's Valve in Open WebUI
- `LINK_SIGNING_KEY`: restart `policy-api`; existing download links stop working
- `LLM_API_KEY`: restart the local model services (escalate, extract), `policy-api` and `open-webui`
- `GRANITE_API_KEY`: change it on the Granite server and in `.env`, then `docker compose up -d granite-gateway`. Nothing else needs it
- Never paste secrets into chats, tickets or a wiki.

Models and supply chain

- Use only models on your organisation's approved list. All defaults are US-developed and published under permissive licenses. Check each model card and license before use.
- Granite 4.2 30B runs on your own vLLM server, outside this stack. Its weights, vLLM version and flags are that server's owner's job. Prefer the built-in `--reasoning-parser nemotron_v3` over the model card's `granite_thinking_parser.py` plugin, which runs code from the model download.
- Download weights outside the environment, record hashes (`scripts/verify_models.sh record`), carry them in, and check them (`verify_models.sh check`) before every start and upgrade.
- Pin every image tag in `.env`. Never use `latest`. Build the policy-api image from a pinned dependency set or your internal mirror.
- vLLM's `--trust-remote-code` (needed for Inkling) runs Python shipped with the model. Review that code and verify the hashes before enabling `escalate`.

Access

- Open WebUI accounts are created by an admin. Sign-up is off after the first admin.
- Everyone with an account can query every group. If some groups must be restricted, run a separate stack per audience, or add per-user group filtering to the tool before go-live.
- The tool sends the asking user's email or name as `requested_by`, which is stored on the run. Tell users their questions are logged.

Human oversight

The system **finds and explains**. It does not approve changes.

- Every "Yes" and "Maybe" goes to a named person, who records a decision (the Review decision column, later the review queue).
- "Maybe" never means "fine". Not-affected policies are listed, with the reason, so reviewers can spot-check them.
- Don't let the tool's output become the record of decision. The decision belongs in your document control system.
- Measure accuracy on real past changes before launch and every quarter (evaluation.md).

Deciding "maybe" sections in chat (`REVIEW_GATE=1`)

The AI model cannot decide. A person types `decide <check number> ...` as their whole message; the Open WebUI upload filter (trusted code) reads it and sends it to policy-api. The check number is the first 8 characters of the check's id, which is only 32 bits. Plainly, what that means:

- It is a convenience, not a password. Someone who guessed or saw a code could decide that check.
- Only signed-in staff can enter it, and only through the chat (the filter holds the service key; people and the model do not).
- It is accepted only as exactly 8 characters (or the full id), and only on the status and review routes. Files, downloads and every other route need the full id.
- For a review it is looked up only among checks that are paused waiting for a decision. A finished or running check cannot be decided, and a code that matches two paused checks is refused.
- Every decision is recorded under the Open WebUI name of whoever typed it, so a wrong or unauthorised one is visible and can be traced.
- The check is not tied to the person who started it. The tool stores the asking person's email or display name as `requested_by`, but that is a label, not a stable user id, so it cannot be matched safely to whoever types the decision. Treat check numbers as internal: do not paste them where people outside the team can see.

Data retention

Uploaded files, runs, verdicts and Excel files stay in the database and the `exports` volume until you delete them. Agree on a retention period with records management. To purge runs older than a set age:

```
-- First, the saved graph state of those runs (it is keyed by run id and is not removed with the
run):
DELETE FROM checkpoint_writes WHERE thread_id IN (SELECT id::text FROM runs WHERE created_at <
now() - interval '2 years');
DELETE FROM checkpoint_blobs  WHERE thread_id IN (SELECT id::text FROM runs WHERE created_at <
now() - interval '2 years');
DELETE FROM checkpoints       WHERE thread_id IN (SELECT id::text FROM runs WHERE created_at <
now() - interval '2 years');
DELETE FROM runs WHERE created_at < now() - interval '2 years';   -- cascades to changes and
judgements
DELETE FROM uploads u WHERE NOT EXISTS (SELECT 1 FROM runs r WHERE r.upload_id = u.id)
                        AND u.created_at < now() - interval '2 years';
```

Then delete the matching `<run-id>.xlsx` files from the `exports` volume.

If something goes wrong

Situation	Do this
`check_containment.sh` fails	Stop `open-webui` and `policy-api` (`docker compose stop`). Fix the network. Re-run the check. Record what happened
A secret may have leaked	Rotate it now (see Secrets). Check the proxy logs for unusual downloads
A wrong result was acted on	Find the run ID on the Excel Summary sheet; pull its `judgements`; add the case to the evaluation set
A policy contains text that tries to instruct the model	Keep the document for analysis; check the verdicts for that policy by hand

Go-live checklist

- [ ] `check_containment.sh` passes and the output is filed
- [ ] Host firewall blocks outbound traffic from the server, except `llm-egress` to the Granite address only
- [ ] `GRANITE_URL` uses `https://` (unless Granite is on this machine) and the certificate check works
- [ ] `.env` secrets generated fresh, file is `chmod 600`, not in git
- [ ] Model hashes verified; models on the approved list
- [ ] Image tags pinned; none is `latest`
- [ ] Open WebUI sign-up turned off; users created by an admin
- [ ] TLS on the reverse proxy in front of `proxy`; `PUBLIC_BASE_URL` uses `https://`
- [ ] Evaluation recall meets the agreed target on a hold-out set
- [ ] Review process for Yes/Maybe agreed with the policy owners
- [ ] Database backups running and restore tested
- [ ] Retention period agreed and written down